

APEX: AI Platform of Platforms

Enterprise Architecture on AWS Agent Core

A comprehensive TOGAF 10 Architecture Development Method (ADM) guide for designing, governing, and delivering an AI Platform of Platforms at NexaBank Global.

Incorporates AWS AI-Driven Development Lifecycle (AI-DLC) impact analysis across all nine TOGAF phases, with enterprise architecture implications, operating model changes, and governance adaptations required for the AI-native enterprise.

ORGANISATION
NexaBank Global

FRAMEWORK
TOGAF 10 ADM

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APEX — AI Platform of Platforms

CLOUD PLATFORM
AWS Agent Core

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OVERVIEW

Executive Summary

Context, scope, and key facts

About This Document

This document is the authoritative Enterprise Architecture (EA) reference for the APEX programme at NexaBank Global. It applies the TOGAF 10 Architecture Development Method (ADM) across all phases to architect, govern, and deliver a unified AI Platform of Platforms on AWS Agent Core. It has been extended to incorporate the impact of the **AWS AI-Driven Development Lifecycle (AI-DLC)**, a methodology introduced by AWS in July 2025 that fundamentally reimagines how enterprises develop software using AI as a central collaborator — not merely an assistant.

AI-DLC has direct and material consequences for enterprise architecture practice. Where traditional TOGAF ADM cycles operate over months, AI-DLC 'bolts' (the AI-DLC equivalent of sprints) operate over hours to days. This document analyses these impacts systematically across every ADM phase and recommends specific adaptations for the NexaBank enterprise architecture operating model.



Business Context — NexaBank Global Pain Chain

Problem	Current State	APEX Target	AI-DLC Multiplier
Time-to-market for AI	9.2 months	2.5 months	AI-DLC 'bolts' → <2 months
Duplicated AI spend	\$14M/year	\$3M/year	Reuse via agent marketplace
DORA governance gaps	47 gaps	0 gaps	AI-generated compliance artefacts
Developer NPS	-12	+55	AI-DLC mob construction
Agents in production	23 (siloeed)	120 (platform)	Self-service + AI-DLC
MTTR for AI incidents	6.3 hours	<30 minutes	AI-DLC ops phase automation

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SECTION 01

AI-DLC: Research Findings & Enterprise Architecture Impacts

AWS AI-Driven Development Lifecycle — Implications for TOGAF 10 Practice

What is AI-DLC?

The **AI-Driven Development Lifecycle (AI-DLC)** was introduced by AWS in July 2025 and presented as a keynote methodology at AWS re:Invent 2025 (session DVT214). It is the result of over 100 enterprise customer experiments by AWS Solutions Architecture and was built on observations that the two existing approaches to AI in software development — 'AI-assisted' (AI helps narrow tasks) and 'AI-autonomous' (AI builds everything) — both produced suboptimal results in velocity and quality. AI-DLC positions AI as a **central collaborator and teammate**, not merely a tool.

Early adopters including Wipro and Dun & Bradstreet reported **10–15x productivity gains**. A team at Wipro that had months of work planned completed all of it in just 20 working hours using AI-DLC methodology. The Atlassian RovoDev 2026 study found that AI agents in code review led to additional fixes in 38.7% of cases, while the Qodo 2025 AI Code Quality Report showed quality improvements rising to 81% from 55% with AI-assisted reviews.

The Three Phases of AI-DLC

Phase	Name	AI Role	Human Role	EA Impact
1	INCEPTION	Transforms business intent into requirements and questions	Validates, stories, and models business requirements	Establishes business context, and requirements
2	CONSTRUCTION	Proposes architecture, domain models, and code	Clarifies, refines, and builds artifacts	Architectures, models, and code artifacts
3	OPERATIONS	Applies accumulated context to manage and improve	Releases and deploys infrastructure	Operational systems and infrastructure

AI-DLC vs Traditional Agile: Terminology Changes

AI-DLC introduces new terminology to reflect its AI-driven, high-intensity collaborative approach. Understanding these changes is important for NexaBank's programme management and architecture governance:

Traditional Agile Term	AI-DLC Term	Duration Change	Architecture Implication
Sprint	Bolt	2–4 weeks → hours to days	ADM phases must operate within bolt cadence
Epic	Unit of Work	Months → days to weeks	Architecture work packages shrink; more frequent releases
User Story	AI-Elaborated Requirement	Days → minutes	Requirements traceability automated
Code Review (PR)	AI-Augmented Quality Scan	Hours → minutes	Automated conformance checking
Architecture Review	Real-time Mob Construction	Weeks → concurrent	ARB must provide real-time guidance
Documentation	Persistent Semantic Context	Post-sprint → continuous	Architecture repository auto-updated
Retrospective	Continuous Learning Loop	End of sprint → continuous	Architecture debt auto-detected

AI-DLC Impact Matrix: All 10 TOGAF ADM Phases

The following matrix maps AI-DLC's structural implications to each TOGAF ADM phase, identifying what changes, what risks emerge, and what adaptations NexaBank must make:

TOGAF Phase	AI-DLC Impact	Risk	NexaBank Adaptation
Preliminary	AI generates architecture principles	Principles quality depends on data and prompts	AI-generated principles feed into all subsequent phases

Phase A — Vision	AI transforms business intent into draft architecture, identifying risks and dependencies.	Stakeholder review and validation of business goals and vision.	Stakeholder review and validation of business goals and vision.
Phase B — Business	AI auto-generates business capability maps, identifying gaps and opportunities.	Existing process mapping and identification of business capabilities.	Identification of business capabilities and their relationships.
Phase C — Data	AI creates logical data models, entity relationships, and data flows.	Information systems, data sources, and data integration requirements.	Information systems, data sources, and data integration requirements.
Phase C — Application	AI proposes application component models and integration patterns.	Application components, integration patterns, and data flows.	Application components, integration patterns, and data flows.
Phase D — Technology	AI generates Terraform IaC, security policies, and deployment scripts.	Technology stack, security policies, and deployment scripts.	Technology stack, security policies, and deployment scripts.
Phase E — Opportunities	AI can analyse gap reports and propose strategic initiatives.	Business process gap analysis and strategic initiative identification.	Business process gap analysis and strategic initiative identification.
Phase F — Migration	Bolts replace work packages; migration plans generated.	Work packages, migration plans, and deployment scripts.	Work packages, migration plans, and deployment scripts.
Phase G — Governance	AI automates conformance checking against standards.	Standards, conformance checks, and audit trails.	Standards, conformance checks, and audit trails.
Phase H — Change	AI monitors technology landscape (new tools, changes, proposals).	Technology landscape, changes, and proposals.	Technology landscape, changes, and proposals.
Req. Management	AI generates and maintains requirements, ensuring traceability.	Requirements, traceability, and change management.	Requirements, traceability, and change management.

Operating Model Transformations Required

AI-DLC is not just a development methodology — it requires structural changes to the enterprise architecture operating model. Research from Intelance, Staun & Stender, and CIO.com (2025) identifies five critical transformation areas that NexaBank must address:

1. Governance Tempo: From Quarterly to Near Real-Time

Traditional EA governance operates on quarterly review cycles. AI-DLC's 'bolt' cadence means architecture decisions are needed in hours. NexaBank must establish a tiered governance model: automated checks run continuously (Cat A/B compliance), architect-in-the-loop reviews occur daily, full ARB convenes weekly rather than bi-weekly. Research from Staun & Stender (2025) confirms that 'these are not quarterly review topics; they are near real-time decisions.'

2. EA Artefacts: From Static Documents to Living Digital Twins

Gartner (2025) notes that Forrester identified AI agents in EA suites are now automating data validation, capability mapping, and artefact creation. NexaBank's architecture repository (LeanIX) must transition from a periodic-update model to a continuously auto-updated environment where AI-DLC tools write back architecture artefacts as they are generated. The EA team shifts from artefact authors to artefact reviewers.

3. New Roles: Enterprise AI Architect & Chief AI Ethics Officer

BCG research found that 89% of C-level executives ranked AI/GenAI in their top three strategic priorities for 2024. In response, organisations are introducing dedicated roles such as the Enterprise AI Architect. NexaBank must create this role (filled internally from the APEX Architecture team) and formalise the Chief AI Ethics Officer position proposed in the APEX Business Architecture.

4. Data Architecture Extension: AI-Native Entities

TOGAF's traditional data architecture covers conceptual, logical, and physical models. AI-DLC introduces new first-class data entities that must be governed: **embeddings** (versioned vector representations), **prompt catalogs** (regulated as code + data), **feedback logs** (agent decision outcomes), and **model lineage records** (EU AI Act traceability). MLflow's Prompt Registry exemplifies this shift in tooling, as identified in the TOGAF BDAT stress-test research (Medium, Sept 2025).

5. Business Architecture: Dynamic Capabilities Replace Static Models

Traditional Business Architecture documents stable capabilities. AI-DLC and agentic AI create capabilities that 'learn and adapt in motion.' Examples: Walmart and Carrefour already use AI to shift pricing and promotions dynamically — what was a quarterly campaign is now an hourly decision. NexaBank's Business Capability Map (BCM-APEX-001) must be updated to flag 'adaptive capabilities' and their governance implications separately from static capabilities.

NexaBank AI-DLC Adoption Roadmap

NexaBank will adopt AI-DLC in a phased approach aligned to APEX delivery horizons, to avoid disrupting in-flight architecture governance while capturing productivity gains:

Horizon	Period	AI-DLC Adoption Scope	EA Impact	Governance Gate
H0: Observe	Q1 2025	AI-DLC training for architecture team, pilot on existing packages	Minimal, existing packages maintained	CTO sign-off on pilot
H1: Adopt for Build	Q2–Q3 2025	AI-DLC Construction phase for Pioneer 1–2, core data flows	Phase 0 Data & Tools Assisted, PE approves and assisted	PE approves and assisted
H2: Extend to Incept	Q4 2025	Mob Elaboration for Pioneer 3–5 requirements and architecture	Phase A/S and architecture reviewed, PE Reg Co-own process	PE Reg Co-own process
H3: Full AI-DLC	Q1–Q2 2026	End-to-end AI-DLC across all APEX Full blueprint, model transformation, Architect X-B integration	Full blueprint, model transformation, Architect X-B integration	Full blueprint, model transformation, Architect X-B integration

PRELIMINARY

Preliminary Phase — Foundation

Establish architecture capability, principles, and governance

The Preliminary Phase establishes NexaBank Global's architecture capability for the APEX programme. Before any AI platform design begins, we codify governance structures, tailor the ADM for financial services regulatory constraints, and ratify the architecture principles that govern every subsequent decision.

Key Activities

- Define Architecture Capability & Maturity Target (CMMI Level 3 target)
- Establish the Architecture Board (ARB) and full RACI
- Tailor the TOGAF ADM for FinServ regulatory constraints (DORA, BCBS 239, EU AI Act)
- Populate the Architecture Repository: LeanIX + Confluence + Azure DevOps
- Define and ratify Architecture Principles (9 principles across 3 domains)
- Establish architecture tooling standards and modelling conventions (ArchiMate 3.2 + C4)
- AI-DLC Impact: Load NexaBank architecture standards as AI-DLC 'steering files' in Amazon Q Developer

Document: Architecture Principles (APD-001)

DOC REF: APD-001 | v2.1 | APPROVED

Business Principles

ID	Principle	Statement	Key Implication
BP-01	AI as Business Capability	Every APEX agent traces to a measurable business outcome in the Corporate Strategy Map (ESM)	Business outcomes must be Corporate Strategy Map (ESM)
BP-02	Human-in-the-Loop by Default	All agents acting on customer data or financial decisions must have a 21-day final human oversight	AWS Bedrock Human-in-the-Loop (HIL) final human oversight
BP-03	Platform Thinking over Point Solutions	Do not procure a standalone AI/ML tool. APEX capability provided via agent AI capabilities within the Cloud	ARBPX can provide equivalent AI capabilities within the Cloud

Data Principles

ID	Principle	Statement	Key Implication
DP-01	Data Sovereignty and Residency	PII and financial data must remain in-region AWS deployment; restrict model endpoints	Multi-region AWS deployment; restrict model endpoints
DP-02	Data as a Shared Asset	All data consumed by APEX agents is cataloged in the NexaBank AWS Data Catalog and .	Cataloged in the NexaBank AWS Data Catalog and .
DP-03	Explainability Before Deployment	No ML model embedded in an APEX agent until clearly documented explainability gates in APEX	Agent Metadata clearly documents explainability gates in APEX
DP-04 (APD-001)	AI Data Catalogs as Governance	Embed Data templates, embeddings, and feature sets into prompt registries plan assets going in City the Data	Multi-Region Prompt Registries plan assets going in City the Data

Technology Principles

ID	Principle	Statement	Key Implication
TP-01	AWS-First, Not AWS-Only	AWS is the strategic cloud provider; architecture must be portable agent	Agent architecture is portable agent
TP-02	Security by Design	Security controls are embedded at design time, not retrofitted post-deployment	Design time controls post-deployment
TP-03	Observability as First-Class	Every APEX agent must emit structured CloudWatch metrics and logs to the data lake; Datadog APM for	CloudWatch metrics and logs to the data lake; Datadog APM for
TP-04	Cost Transparency	Each AI agent must have a tagged cost center and monthly tags for machine learning ops review gate	AWS Cost and monthly tags for machine learning ops review gate
TP-05 (AI-DLC)	Dev Understands AI	All generated IaC and code must be reviewed and understood by AI-generated generators for onk	Reviewed and understood by AI-generated generators for onk

PHASE A

Architecture Vision

Define the why, what, and who — secure executive mandate

Phase A creates the Architecture Vision for APEX: the compelling case for change, stakeholder landscape, architecture scope, high-level target state, and the Statement of Architecture Work (SAW) that gives the programme formal mandate. AI-DLC Mob Elaboration sessions accelerate this phase significantly.

Document: Statement of Architecture Work (SAW-APEX-001)

DOC REF: SAW-APEX-001 | v1.4 | APPROVED – Group CTO sign-off: 2025-02-14

Architecture Vision Statement

"By Q4 2026, NexaBank Global will operate a unified AI Agent Platform (APEX) on AWS Agent Core that reduces time-to-market for AI use cases from 9.2 to 2.5 months, achieves full EU AI Act and DORA compliance, and generates £38M NPV over 5 years through platform economics."

Architecture Constraints

Constraint	Source	Architecture Impact
Data must not leave contractual jurisdiction	GDPR / PDPA / LGPD	Multi-region AWS deployment mandatory
All AI models require Model Risk sign-off	SR 11-7 / Internal Policy	6-week validation gate per model class
AWS is strategic cloud partner	Board Cloud Strategy	Primary platform must be AWS; abstraction layers required
Zero-downtime for Tier-1 services	SLA Policy	Blue-green deployment mandatory for all Pioneer agents
99.95% availability target	Enterprise SLA	Multi-AZ + DR across 2 regions minimum
EU AI Act enforcement from Aug 2026	European Commission	High-Risk AI classification and controls pre-built

Stakeholder Power/Interest Grid

Key Players (High Power, High Interest) — Core governance engagement:

Stakeholder	Role	Primary Concerns	APEX Value Proposition	Engagement Cadence
Sarah Chen	Group CTO / Executive Sponsor	Spex, cost, regulatory risk	68% TTM; -\$14M duplicated spend	Monthly steering
Marcus Webb	VP AI Platform	Platform adoption, developer experience	Self-service marketplace	Daily collaboration
Priya Nair	CISO	Data leakage, model poisoning	Centralised Bedrock Guardrails + ABAC	Bi-weekly review
James Okafor	Group Compliance	DORA Art.11, EU AI Act Art.24	Automated compliance reporting	Monthly evidence
Dr. Lin Wei	Chief Data Officer	Data mesh integration, lineage, governance	Native, Cloud-Agnostic integration	Bi-weekly
PRA / ECB	Regulator	Systemic AI risk, model governance	DORA evidence pack, EU AI Act Art.24	Quarterly briefing

Five Pioneer Domains (Proof-of-Value)

#	Domain	Agent Name	Use Case	Key Regulation	AI-DLC Bolt Estimate
1	Retail Banking	CreditRiskAgent	Automated credit decisioning with explainability	SR 11-7, FCA	3 bolts (approx. 6 weeks)
2	Corporate Banking	KYCRerefreshAgent	Continuous KYC monitoring, sanctions screening	FDICIS, SAR, FATF	3 bolts
3	Wealth Management	PortfolioRebalAgent	Rule-based rebalancing recommendations	SEC Rule 27	2 bolts
4	IT Operations	IncidentAgent	L1/L2 ticket triage and remediation	DORA Annex	2 bolts
5	Risk	ModelMonitorAgent	Drift detection, backtesting, regulatory reporting	SR 11-7, ECB, PRA	3 bolts

PHASE B

Business Architecture

Map capabilities, value streams, and organisational model

Phase B produces the Business Architecture for APEX — the capability model, value streams, organisation design, and business processes that the platform must support. AI-DLC fundamentally changes how business capabilities are modelled: capabilities that use AI are no longer static — they 'learn and adapt in motion.'

Business Capability Map — Gap Summary (BCM-APEX-001)

DOC REF: BCM-APEX-001 | v1.5

L1 Capability Domain	L2 Count	Critical Gaps	APEX Addresses	AI-DLC Dynamic?
1. AI Strategy & Governance	5	4 critical	5/5	Partially — policy enforcement via AI ag
2. Agent Development	5	5 critical	5/5	YES — AI-DLC Construction phase
3. Agent Orchestration	4	4 critical	4/4	YES — AI-DLC Operations phase
4. Knowledge Management	4	4 critical	4/4	YES — continuous KB refresh
5. AI Operations (AIOps)	5	5 critical	5/5	YES — AI-DLC Operations + H-phase
6. Data Integration	4	2 critical	3/4	PARTIALLY — Data Mesh remains dom
7. Platform Enablement	4	4 critical	4/4	YES — self-service portal auto-generate

Value Stream: AI Agent Delivery (Baseline vs APEX vs AI-DLC)

DOC REF: VSA-APEX-001 | v1.1

Stage	Baseline Duration	APEX Duration	AI-DLC Duration	AI-DLC Enabler
Intake & Qualification	2–4 weeks	5 business days	1–2 hours	AI-DLC Mob Elaboration; EU AI Act classif
Architecture & Design	4–8 weeks	10 business days	1–3 days (1 bolt)	Mob Construction; AI generates architectu
Build & Integrate	8–16 weeks	3 weeks	1–2 weeks (1–2 bolts)	Amazon Q Developer; AI generates IaC, L
Validate & Govern	6–10 weeks	10 business days	3–5 days	AI-automated RAGAS eval, Guardrail tests
Deploy & Monitor	2–4 weeks	1 week	1–2 days	AI-DLC Operations; IaC apply, canary, das
TOTAL TIME-TO-MARKET	22–42 weeks (~9mo)	9.5 weeks (~2.5mo)	3–4 weeks (~1mo)	AI-DLC compounds across all stages

Value Waste Elimination

Waste Type (Lean)	Current Cost	APEX Reduction	AI-DLC Further Reduction	Residual
Rework (duplicate agent patterns)	\$3.2M	-80% via platform reuse	-15% via AI pattern suggestions	\$0.2M
Waiting (manual compliance queues)	\$4.8M	-60% via automated pipelines	-30% via AI-generated evidence	\$0.5M
Overprocessing (obs stack rebuilds)	\$2.1M	-90% via shared observability	-5% via AI-generated dashboards	\$0.1M
Inventory (abandoned agent builds)	\$1.9M	-70% via marketplace discipline	-20% via AI-DLC fast completion	\$0.3M
Defects (untested agents in prod)	\$2.0M	-75% via CI/CD gate	-20% via AI-generated test suites	\$0.1M
TOTAL	\$14.0M/yr	\$7.6M reduction	\$1.5M additional reduction	\$1.1M residual

PHASE C

Information Systems Architecture

Data architecture and application component design

Phase C covers both Data and Application Architecture. For APEX, this is the most technically complex phase: it introduces AI-native data entities (embeddings, vector stores, prompt catalogs, feedback logs) that do not exist in traditional TOGAF data models, and agent-oriented application design patterns that require new architectural conventions.

Data Classification Framework (APEX-Specific)

DOC REF: DAD-APEX-001 | v1.3

Class	Description	Examples	Bedrock Guardrails Action	Retention
C4 — Highly Restricted	PII + Financial data	NI numbers, account balances	EN, ENit, log, re-tokenise in prompts	7 years (DORA)
C3 — Confidential	Internal business data	Credit models, risk parameters	Re-encrypt and logs	5 years
C2 — Internal	Non-sensitive operational	Agent performance metrics, State logs	State logging	3 years
C1 — Public	Regulatory/market data	FCA rules, market rates, Base UI guides	State logging	Indefinite
C-AI — AI Assets (NEW)	AI-DLC: Prompt catalogs, Embeddings, Feedback logs, RAG logs, AI O sign-off logs	Prompt catalogs, Embeddings, Feedback logs, RAG logs, AI O sign-off logs	AI O sign-off logs	Indefinite

AI-Native Data Entities (AI-DLC Extension to TOGAF Data Architecture)

Traditional TOGAF data architecture does not include AI-specific artefacts. AI-DLC and agentic platforms introduce a new category of governed data entities that must be treated as first-class architectural concerns:

Entity	Description	Governance Owner	Tooling	Risk if Ungoverned
Prompt Catalog	Versioned prompt templates	Platform Engineering	AI O sign-off logs	Regulatory non-compliance
Embedding Store	Vector representations of knowledge	Platform Engineering	AI O sign-off logs	Regulatory non-compliance
Feedback Log	Human-in-Loop decisions, model risks	Platform Engineering	AI O sign-off logs	Regulatory non-compliance
Agent Memory	Cross-session context store	Platform Engineering	AI O sign-off logs	Regulatory non-compliance
Model Lineage Record	Training data → model version	Platform Engineering	AI O sign-off logs	Regulatory non-compliance
RAG Retrieval Log	What chunks were retrieved	Platform Engineering	AI O sign-off logs	Regulatory non-compliance

Knowledge Base Architecture (RAG Design)

KB ID	Name	Sources	Chunking	Embedding Model	Refresh	Vector Store
KB-001	Credit Policy KB	FCA Handbook (2,847pp), NexaBank C12 Policy	Recursive splitting	Amazon Bedrock (Claude 3.5)	Weekly (regulatory updates)	Amazon Bedrock
KB-002	KYC Regulatory KB	FinCEN guidance, FCA AMIS handbook	Recursive splitting	Amazon Bedrock (Claude 3.5)	Weekly (regulatory updates)	Amazon Bedrock
KB-003	Product & Suitability KB	NexaBank product catalogue	Recursive splitting	Amazon Bedrock (Claude 3.5)	Weekly (regulatory updates)	Amazon Bedrock
KB-004	IT Runbook KB	ServiceNow runbooks, DORA guidance	Recursive splitting	Amazon Bedrock (Claude 3.5)	Weekly (regulatory updates)	Amazon Bedrock

Application Architecture — AWS Agent Core Component Model

DOC REF: AAD-APEX-001 | v1.3

Component	Technology	Pattern	SLA	AI-DLC Construction Note
APEX Developer Portal	React SPA + API Gateway	BFF (Backend-for-Frontend)	99.9% / p99 <500ms	UI scaffolded by Amazon Q Developer
APEX Control Plane	Python FastAPI on ECS Fargate	Event-driven service (5 serverless)	99.9% / p99 <500ms	FastAPI skeleton + unit tests AI-generated

Bedrock Supervisor Agent	AWS Bedrock Agents + Claude 3.5	Orchestrator-Worker	99.95% / p99 <3s	Agent YAML + prompt templates generated
Tool Lambda Library	AWS Lambda Python 3.12	Sidcar + Action Groups	99.95% / p99 <1s	Each tool function AI-generated; mandatory
Human Loop (A2I)	AWS Augmented AI + React	Human-in-the-Loop	99.5% / SLA by task type	Workflow config generated; custom
Observability Stack	CloudWatch + X-Ray + DataDog	Centralised OTEL collector	99.5%	Dashboard-as-code generated by AI-DLC
Compliance Reporter	Python + AWS QuickSight	Batch report generation	99.5% / daily run	EU AI Act and DORA report templates

API Catalogue

API	Version	Pattern	Auth Method	Availability SLA	Owner
Agent Registry API	v2	REST/JSON + OpenAPI 3.1	OAuth2 (APEX scope)	99.9% / p99 <200ms	Platform Eng
Agent Invocation API	v1	REST + SSE streaming	mTLS SigV4	99.95% / p99 <3s	Platform Eng
Knowledge Base API	v1	REST/JSON	IAM SigV4	99.9% / p99 <500ms	AI/ML Eng
Onboarding API	v1	REST/JSON	OAuth2	99.5% / p99 <1s	Platform Eng
Chargeback API	v1	REST/JSON	mTLS	99.9% monthly	FinOps
Compliance Report API v1		REST/JSON	OAuth2 + MFA	99.5% daily	Compliance

PHASE D

Technology Architecture

Multi-region AWS infrastructure, security, CI/CD, and observability

Phase D specifies the complete technology architecture for APEX on AWS. This covers multi-region infrastructure, AWS Agent Core configuration, network design, security architecture, the CI/CD pipeline enhanced with AI-DLC tooling, and the operational observability model.

Multi-Region Deployment Model

DOC REF: TAD-APEX-001 | v2.0

AWS Region	Role	Data Types	Key Services	Regulatory Basis
eu-west-1 (Ireland)	PRIMARY	UK/EU customer PII, financial data	Bedrock Agents, AOSS, Aurora, S3, OpenSearch	GDPR, FCA, PRA
eu-central-1 (Frankfurt)	DR + DE data	German banking data, S3 CloudTrail	Bedrock AOSS (DR), S3	GDPR, BaFin MaRisk
us-east-1 (N. Virginia)	US INSTITUTIONAL	US Institutional counterparty data	Bedrock, AOSS, Aurora replica	CCPA, FinCEN, SEC, FINRA
ap-southeast-1 (Singapore)	APAC	Singapore, HK customer data	Bedrock, OpenSearch Service, S3	PDPA, MAS TRM Guidelines

AWS Bedrock Agent Core Configuration

The APEX Supervisor Agent is configured as an orchestrator that routes to specialist sub-agents per Pioneer Domain. Key configuration parameters:

Parameter	Value	Rationale
Foundation Model	anthropic.claude-3-5-sonnet-20241022-v2:0	Best reasoning performance on NexaBank FS eval suite
Idle Session TTL	1800 seconds (30 min)	Balance UX continuity with PII retention risk; GDPR data minimization
Memory Type	SESSION_SUMMARY (enabled)	Cross-turn context for multi-step financial workflows
Guardrail Version	apex-guardrail-prod v3	PII ANONYMIZE for SORT_CODE, ACCOUNT_NUMBER
Grounding Threshold	0.75 (strict)	Reject responses not grounded in retrieved context — DORA
Max Tokens (output)	4,096	Sufficient for credit decision explanations; cost-control
Invocation Endpoint	VPC PrivateLink only	No public internet path; DORA and security architecture
Bedrock Tracing	Enabled → CloudWatch	DORA Art.11 audit trail; EU AI Act traceability requirement

Security Architecture Summary

Security Domain	Control	Implementation	Compliance Basis
Identity & Access	AWS SSO + NexaBank IdP (SAML)	Centralized Identity Centre; TOTP mandatory	DORA Annex 1, ISO 27001
Encryption at Rest	KMS Customer Managed Keys (CMK)	Separate CMK per data class; 365-day rotation	DORA Art. 32, PCI DSS
Encryption in Transit	TLS 1.3 minimum	TLS 1.0/1.1 disabled via AWS policy; FIPS 140-2 DRBs	FINMA, DORA
Network Isolation	VPC PrivateLink for all Bedrock calls	No public internet from Lambda/ECS; VPC Security Groups	SPD, Security Policy
Threat Detection	GuardDuty + Macie + Security Hub	All Regions; S3 PII scanning daily; findings auto-remediate	DORA Art. 31, ISO 27002
Agent Safety	Bedrock Guardrails v3	PII anonymisation; financial fraud filter; EU data handling	GDPR Art. 32, DORA
Prompt Injection Defence	Input sanitisation Lambda wrapper	All agent inputs pass through sanitisation before Bedrock	OWASP LLM Top 10
Secrets Management	AWS Secrets Manager	90-day auto-rotation; no hardcoded secrets	ISO 27001, AWS (S2 checkov enforced)

CI/CD Pipeline — AI-DLC Enhanced

The APEX CI/CD pipeline has been redesigned to incorporate AI-DLC Construction and Operations phase tooling alongside traditional quality gates:

Stage	Tools	AI-DLC Enhancement	Gate Type
1. Code Generation	Amazon Q Developer, GitHub Copilot	AI generates Terraform IaC, Lambda function code	Human Review Mandatory (EP-05)
2. Static Analysis	Checkov, cfn-lint, yamllint, OWASP	AI scans any failures with remediation suggestions	Automated; zero HIGH/CRITICAL to pass
3. Prompt Regression	RAGAS (custom eval suite)	AI-generated test cases against baseline	>90% score required on regression
4. Guardrail Testing	Bedrock Guardrail test harness	AI generates adversarial prompts to test guardrails	Automated; 100% pass required
5. Model Risk Gate	Azure DevOps approval workflow	AI generates model risk evidence pack draft	Human Model Risk sign-off
6. Non-Prod Deploy	Terraform + GitHub Actions	AI/DC Operations: IaC apply automated; smoke tests AI-generated	Automated; AI-generated review
7. Canary Deploy	AWS CodeDeploy (10%→50%→100%)	AI monitors canary metrics; auto-rollback on 400+ errors	AI-led with human override
8. Post-Deploy	CloudWatch + FinOps checks	AI generates stakeholder summary; cost tags auto-applied	Automated; DRP evidence created

PHASES E–H

Opportunities, Migration, Governance & Change

Roadmap, delivery plan, compliance, and ongoing architecture management

Phase E — Architecture Roadmap (ROAD-APEX-001)

DOC REF: ROAD-APEX-001 | v2.1

AI-DLC compresses the architecture roadmap significantly. Traditional work packages measured in months are replaced by 'bolts' (hours to days). The following roadmap shows both APEX work packages and their AI-DLC bolt estimates:

Work Package	WP-ID	AI-DLC Bolts	Calendar Weeks	Horizon
APEX Core Infrastructure (multi-region VPC, IAM, KMS)	WP-001	2 bolts	4 weeks	H1
Bedrock Agent Core Setup (supervisor, guardrails, prompts)	WP-002	2 bolts	4 weeks	H1
CI/CD Pipeline with AI-DLC tooling	WP-004	1 bolt	2 weeks	H1
Security Baseline (all controls)	WP-008	2 bolts	4 weeks	H1
Observability Stack (CloudWatch + Datadog)	WP-007	1 bolt	2 weeks	H1
Pioneer 1: Credit Risk Agent	WP-005	3 bolts	8 weeks	H1
Pioneer 2: KYC Refresh Agent	WP-006	3 bolts	8 weeks	H1
Pioneer 3: Portfolio Rebalancing Agent	WP-009	2 bolts	6 weeks	H2
Pioneer 4: IT Incident Response Agent	WP-010	2 bolts	5 weeks	H2
Pioneer 5: Model Monitoring Agent	WP-011	3 bolts	6 weeks	H2
Self-Service Onboarding Portal	WP-012	3 bolts	8 weeks	H2
EU AI Act Compliance Pack (automation)	WP-015	2 bolts	5 weeks	H2

Solution Building Blocks — Build/Buy/Reuse Summary (SBB-APEX-001)

DOC REF: SBB-APEX-001 | v1.2

SBB	Decision	Solution	AI-DLC Role
Agent Orchestration Engine	BUY (SaaS)	AWS Bedrock Agents	AI-DLC generates agent YAML and action group configuration
Foundation Models	BUY (SaaS)	Anthropic Claude 3.5 via Bedrock	AI-DLC: AI selects model per task in Construction phase
Vector Store	BUY (Managed)	AWS OpenSearch Serverless	AI-DLC generates index mappings and query patterns
Agent Control Plane	BUILD	Python FastAPI on ECS Fargate	AI-DLC Construction builds entire service in 2 bolts
Developer Portal	BUILD	React + AWS Amplify	AI-DLC UI scaffolding in Mob Construction (1 bolt)
Human Loop UI	BUY + EXTEND	AWS A2I + custom React	AI-DLC generates A2I workflow JSON; custom React for review
Observability	REUSE + EXTEND	Datadog + CloudWatch	AI-DLC generates dashboards-as-code from agent scaffolding
Tool Lambda Library	BUILD	AWS Lambda Python 3.12	AI-DLC generates each tool function; mandatory human review
Compliance Reporter	BUILD	Python + QuickSight	AI-DLC generates EU AI Act and DORA report templates

Phase F — Implementation & Migration (IMP-APEX-001)

DOC REF: IMP-APEX-001 | v1.5

Migration Strategy: 23 Existing AI Tools

Category	Count	Action	AI-DLC Migration Path	Target Date
ADOPT — Already AWS; add APEX governance layer in APEX tool framework; AI-DLC Operations	8 tools	Wrap in APEX tool framework; AI-DLC Operations	AI-DLC Operations: IaC wrapping in Q3 2025	Q3 2025
ADAPT — Works well; needs APEX integration; Expose as APEX tool; APEX Bedrock adapter Lambda in MQL	6 tools	Expose as APEX tool; APEX Bedrock adapter Lambda in MQL	AI-DLC builds adapter Lambda in MQL	Q4 2025
REPLACE — Duplicates APEX capabilities	7 tools	Pioneer agent replaces; decommission legacy	Parallel migration; canary cutover; AI-DLC metrics	Q1 2026
RETIRE — Unused/low value	2 tools	Immediate decommission; add APEX capabilities	AI-DLC Operations: IaC teardown + Q2 2025 generation	Q2 2025

Phase G — Architecture Governance (CAF-APEX-001 + AC-APEX-001)

DOC REF: CAF-APEX-001 | v1.1 – AI-DLC Impact: Real-Time Automated Conformance

AI-DLC fundamentally changes the governance model: where traditional conformance reviews occur at milestone gates, AI-DLC enables **continuous automated conformance**. The APEX compliance assessment framework operates in two tiers:

Tier	Checks	Method	Frequency	Human Involvement
Tier 1: Automated (AWS Config rules, Checkov, SecurityHub, PACIAS scans, AWS Security Compliance)	Automated	QHOD, PACIAS scans, AWS Security Compliance	Continuous	AI only; human reviews find
Tier 2: Architect-in-Loop	Architectural conformance (design to blueprint), change stage	Automated, change stage	Event-driven / low rate	Daily digital check and signs off
Tier 3: ARB Review	Significant deviations, new SBBs, regulatory ARB findings (and weekly reviews)	Regulatory ARB findings (and weekly reviews)	Weekly (via AI-DLC) / ARB	ARB dated for AI-DLC audit
Tier 4: Steering Committee	Strategic deviations, budget impacts	Executive steering group (AI-generated draft, human reviewed)	Monthly	Steering committee; executive

Phase H — Architecture Change Management (ACHG-APEX-001)

DOC REF: ACHG-APEX-001 | v1.2 – AI-DLC Impact: Continuous Change Intelligence

With AI-DLC and AWS Bedrock model releases occurring frequently (Claude versions, new Bedrock features), Phase H must operate as a continuous intelligence function — not a periodic review. NexaBank implements an AI-powered change monitoring system that watches technology and regulatory landscapes:

Change Type	AI-DLC Monitoring Method	Response SLA	Example
Type 1: Tech-driven (AWS/AQ Deploy) + Bedrock model announcements	AI/ML Ops (Deploy) + Bedrock model announcements	24 hours	Change AWS SageMaker. Bedrock. 23% FS eval i
Type 2: Regulatory-driven	EU AI Act enforcement calendar monitoring	10 business days	EU AI Act enforcement date: APEX comp
Type 3: Business-driven	Domain team agent onboarding request	5 business days	Onboarding to APEX market
Type 4: Simplification	APEX usage analytics identifies underutilized agents	30 days	Agent v1.2 superseded by v2.0; v1.2

REQ MGMT

Requirements Management (Continuous)

Living requirements — the heartbeat of the APEX ADM

Requirements Management is the continuous, central process of the TOGAF ADM wheel. AI-DLC's 'Semantic Context Building' and persistent context storage transform this from a periodic document into a living, AI-maintained traceability system. NexaBank maintains 147 architecture requirements in the ARS; a representative sample is shown below.

Architecture Requirements Specification — Sample (ARS-APEX-001)

DOC REF: ARS-APEX-001 | v2.3 | Living Document – 147 total requirements

Req ID	Category	Statement (Summary)	Priority	Source	Status
BR-001	Business	Platform supports ≥120 concurrent agent interactions	Must	Capacity model	In Design
BR-003	Business	All agent decisions logged and queryable for 7 years	Must	minimum DORA Art.11	In Design
BR-006	Business	EU AI Act risk tier auto-classified at agent intake	Must	EU AI Act	In Design
DR-001	Data	Customer PII never persisted in agent conversations	Must	GDPR Art.5	In Design
DR-NEW-01	Data (AI-DLC)	Prompt templates versioned in GitHub; changes trigger regression tests	Must	ISO 9001, AISLC	Backlog
DR-NEW-02	Data (AI-DLC)	Embeddings versioned per KB refresh; old versions retained	Must	DORA Art.17, EU AI Act	Backlog
AR-001	Application	Agent Control Plane API: p99 response <500ms	Must	SLA Policy	In Design
TR-001	Technology	All Bedrock invocations via VPC PrivateLink only	Must	public security Policy	In Design
TR-NEW-01	Technology (AI-DLC)	AI-generated IaC peer-reviewed by human engineer before prod	Must	ISO 9001, AI-DLC	In Design
SR-003	Security	Red-team adversarial testing of each agent before production	Must	AI Risk Policy	In Review
CR-001	Compliance	EU AI Act risk classification documented for all agents	Must	EU AI Act Art.9	In Design
CR-002	Compliance	Model explainability method documented (SHAP/LIME) before production	Must	EU AI Act Art.13	In Design
CR-003	Compliance	DORA 72-hour ICT incident notification process implemented	Must	DORA Art.17	In Review
CR-NEW-01	Compliance (AI-DLC)	AI-generated architecture artefacts marked with AISLC metadata	Should	AI Governance	Backlog

APEX compliance posture across all applicable regulatory frameworks

Regulation	Key Articles / Rules	APEX Architecture Control	Evidence Artefact
EU AI Act 2024	Art.6 (High-Risk classification), Art.9 (Risk Management System)	Art.18 (Ker-Space Policy) & Art.20 (Data Protection)	AI Model Cards, Data Lineage Logs, Risk Assessment Reports
DORA (EU) 2025	Art.9 (ICT security), Art.11 (ICT risk management), Art.17 (Incident response)	Art.28 (IT Governance) & Art.30 (Cyber resilience)	IT Security Policies, Incident Response Plans, Cyber Resilience Frameworks
GDPR / UK GDPR	Art.5 (data minimisation), Art.25 (privacy by design), Art.32 (security)	Art.39 (Data Protection Officer role) & Art.44 (Data Protection Impact Assessment)	Data Processing Agreements, Privacy Notices, DPIA Documents
SR 11-7 (Fed Reserve)	Model risk management: validation, documentation, monitoring	Model Risk Rating & Approval Process	Model Validation Reports, Documentation, Monitoring Dashboards
BCBS 239	Data lineage, accuracy, completeness	WSR Global Data Catalogue Integration	Data Lineage Diagrams, Data Quality Metrics, Audit Trails
MiFID II	Art.27 (best execution), Suitability assessment for investment advice	Advised Investment Services (AIS) & MiFID II Disclosure Requirements	Investment Advice Records, Client Suitability Assessments, Disclosure Statements
MAS TRM (Singapore)	Technology Risk Management — AIA Model Governance (planned, at least-1); MAS-Defined framework for sovereign Regid		

APPENDIX B

Glossary

Key terms — TOGAF 10, AI-DLC, AWS Agent Core, and regulatory

ADM	Architecture Development Method — the iterative process at the core of TOGAF 10
AI-DLC	AI-Driven Development Lifecycle — AWS methodology (July 2025) positioning AI as central collaborator in software development
APEX	AI Platform of Platforms — the NexaBank programme to build a shared AI agent infrastructure on AWS Agent Core
ARB	Architecture Review Board — governance body that approves architecture decisions and manages conformance
AWS Agent Core	AWS managed service suite including Bedrock Agents, Knowledge Bases, Guardrails, and Flows for building AI agents
Bolt	AI-DLC term for a development cycle; equivalent to an Agile sprint but measured in hours to days, not weeks
DORA	EU Digital Operational Resilience Act — regulation requiring ICT risk management, incident reporting, and TLPT for financial entities
Embedding	Vector representation of text/data used in RAG systems; a first-class data entity in AI-DLC architecture
EU AI Act	European AI regulation (enforcement 2025–2026) classifying AI systems by risk level and mandating transparency, governance, and human oversight
Guardrails	AWS Bedrock feature providing content filtering, PII anonymisation, topic restrictions, and grounding checks for AI agents
Human Loop (A2I)	AWS Augmented AI service for routing AI decisions to human reviewers when confidence is low or policy thresholds are exceeded
KB (Knowledge Base)	AWS Bedrock managed RAG store: documents are chunked, embedded, and stored in a vector database for agent retrieval
Mob Construction	AI-DLC ritual where the entire team works together in real-time with AI to build solutions (replaces isolated coding)
Mob Elaboration	AI-DLC ritual where the team collaborates with AI to transform business intent into requirements and architecture (replaces separate planning sessions)
Pioneer Domain	One of five NexaBank business domains selected as proof-of-value implementations for the APEX platform
Prompt Catalog	Versioned library of prompt templates used by APEX agents; treated as a governed data and code artefact
RAG	Retrieval-Augmented Generation — technique where AI retrieves relevant context from a knowledge base before generating a response
RAGAS	Retrieval-Augmented Generation Assessment — evaluation framework for measuring RAG pipeline quality (faithfulness, relevance, context recall)

SAW	Statement of Architecture Work — TOGAF document that formally initiates an architecture project with scope, constraints, and sponsor approval
SBB	Solution Building Block — a specific implementation of an Architecture Building Block (ABB); defines what will be built/bought/reused
Semantic Context	AI-DLC concept: rich, structured context provided to AI agents through steering files, so they apply organisation-specific standards consistently
TLPT	Threat-Led Penetration Testing — DORA Art.26 obligation for major financial entities to conduct adversarial testing of critical ICT systems
Unit of Work	AI-DLC equivalent of an Agile Epic; a significant deliverable decomposed into bolts; typically days to weeks in duration

Document Control

Version	Date	Author	Change Summary	Approver
1.0	2025-01-15	Enterprise Architect	Initial draft — Phases A–D	ARB
2.0	2025-03-01	Enterprise Architect	Full ADM coverage; Architecture Contracts added	Group CTO
3.0	2026-04-01	Enterprise Architect	AI-DLC impact analysis integrated across all phases; new requirements added	Group CTO

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